PYTHON ABL CASE STUDIES

Program Output Report

Outputs for all 10 case studies

Consolidated Output Document

Case Study 1 — Program Output

Recreated terminal-style output for report presentation

Python Output — Case Study 1

```
>>> python case_study_1.py
===== SCHOLARSHIP RESULTS =====
Rank: 1
Name: Priya
Aggregate: 92.0 %
Subject Average: 93.0 %
Status: Scholarship Allocated
Scholarship Amount: Rs. 20000
Rank: 2
Name: Ravi
Aggregate: 88.0 %
Subject Average: 86.0 %
Status: Scholarship Allocated
Scholarship Amount: Rs. 15000
Rank: 3
Name: Arjun
Aggregate: 80.0 %
Subject Average: 75.0 %
Status: Waitlisted
Total Eligible Applicants: 3
Scholarship Seats Available: 2
```

Case Study 2 — Program Output

Recreated terminal-style output for report presentation

Python Output — Case Study 2

```
>>> python case_study_2.py
Month 1: 85 units, energy I170.00, bill I208.50
Month 2: 120 units, energy I260.00, bill I323.00
Month 3: 190 units, energy I470.00, bill I543.50
Month 4: 310 units, energy I1050.00, bill I1152.50
MONTH-TO-MONTH CHECK
Month 2: unusual increase of 54.9%
Month 3: unusual increase of 68.3%
Month 4: unusual increase of 112.1%
```

Case Study 3 — Program Output

Recreated terminal-style output for report presentation

Python Output — Case Study 3

```
>>> python case_study_3.py
EMPLOYEE SALARY STATEMENT
Employee: Ravi
Basic pay: I24000.00
HRA: I4800.00
Travel allowance: I2400.00
Overtime pay: I1200.00
Incentive: I1500.00
Gross salary: I33900.00
Deduction: I2390.00
Net salary: I31510.00
```

Case Study 4 — Program Output

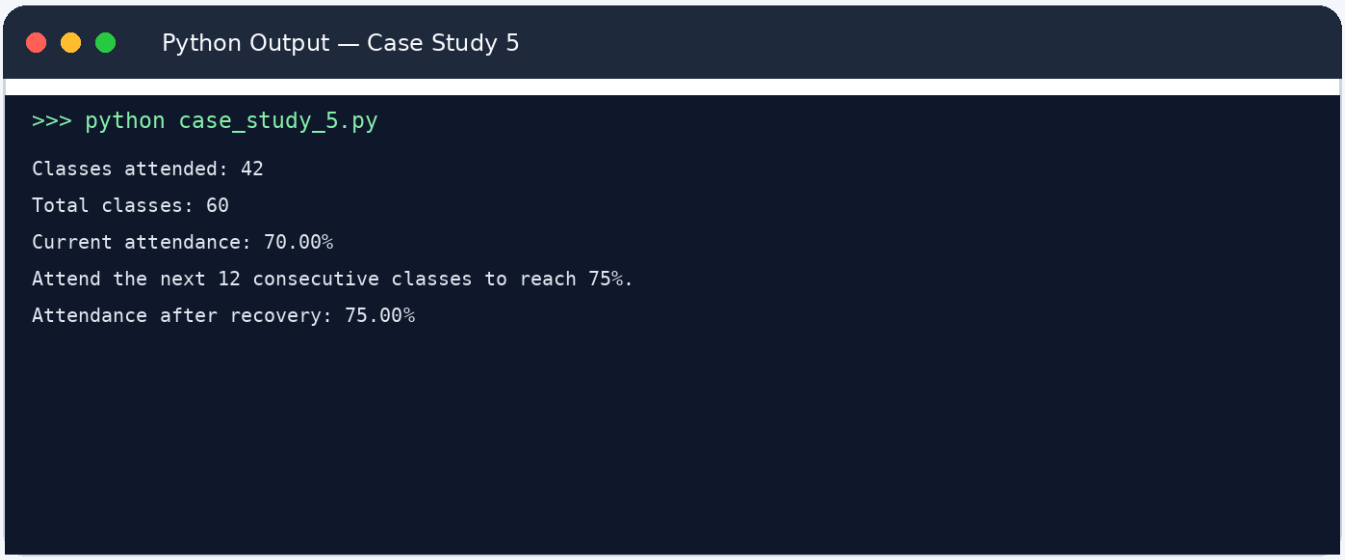
Recreated terminal-style output for report presentation

Python Output — Case Study 4

```
>>> python case_study_4.py
ITEMIZED INVOICE
Notebook: 5 x I80 = I400.00; tax I18.00
School Bag: 2 x I900 = I1800.00; tax I194.40
Pen Pack: 3 x I120 = I360.00; tax I16.20
Subtotal: I2560.00
Discount: I256.00
Tax: I228.60
FINAL AMOUNT: I2532.60
```

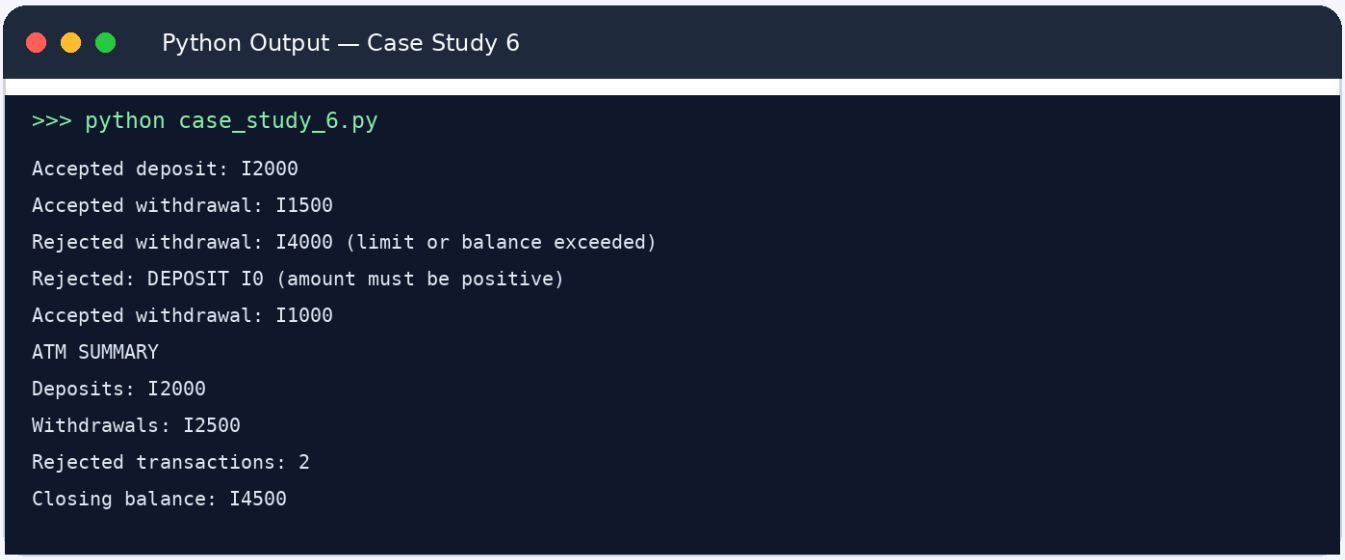
Case Study 5 — Program Output

Recreated terminal-style output for report presentation



Case Study 6 — Program Output

Recreated terminal-style output for report presentation



Case Study 7 — Program Output

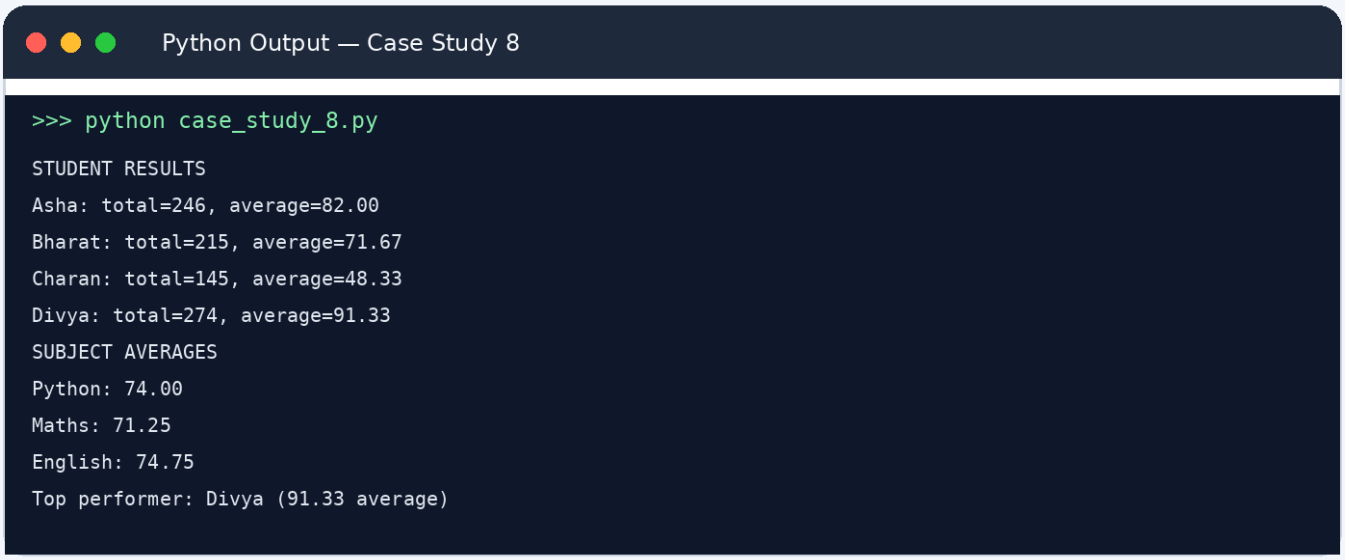
Recreated terminal-style output for report presentation

Python Output — Case Study 7

```
>>> python case_study_7.py
INVENTORY RECONCILIATION
Product: Pens
Expected closing stock: 35
Recorded closing stock: 35
Status: Matched
Replenishment required: 0 units
Product: Notebooks
Expected closing stock: 30
Recorded closing stock: 25
Status: DISCREPANCY (-5 units)
Replenishment required: 0 units
Product: Folders
Expected closing stock: 25
Recorded closing stock: 28
Status: DISCREPANCY (+3 units)
```

Case Study 8 — Program Output

Recreated terminal-style output for report presentation



Case Study 9 — Program Output

Recreated terminal-style output for report presentation

Python Output — Case Study 9

```
>>> python case_study_9.py
Password: abc
Result: Invalid Password
- Password must contain at least 8 characters
- Missing uppercase letter
- Missing digit
- Missing special character
Password: Password123!
Result: Invalid Password
- Contains forbidden text: password
Password: Abc111!xyz
Result: Invalid Password
- Same character repeated 3 times consecutively
Password: Strong@42
Result: Valid Password
Password: Admin@1234
Result: Invalid Password
```

Case Study 10 — Program Output

Recreated terminal-style output for report presentation

Python Output — Case Study 10

```
>>> python case_study_10.py
CAMPUS EVENT BUDGET
Fixed costs: I25000
Variable cost per participant: I180
Expected participants: 100
Total expected cost: I43000
Sponsorship: I18000
Registration contributions: I25000
Total funds: I43000
Funding surplus: I0
Break-even registrations with current sponsorship: 100
```